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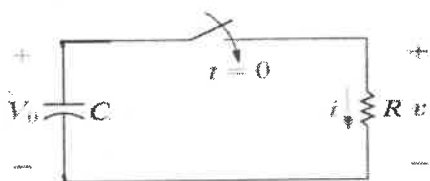
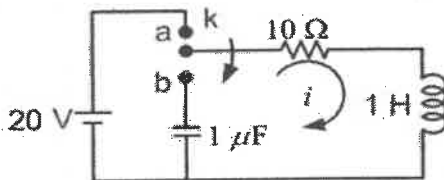
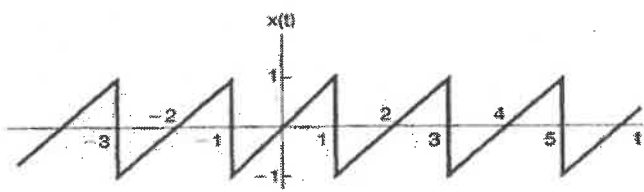
Degree: B.Tech Semester: II**END-SEMESTER EXAMINATION, APRIL-MAY 2026**Course Code: EEEEEC03Course Title: Signal and Systems

Duration: 3 Hours

Max. Marks: 50

**Note:** - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

| Q. No.    | Question                                                                                                                                                                                                                                                       | Marks | COs |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|
| <b>Q1</b> | <b>Attempt any 2 parts of the following</b>                                                                                                                                                                                                                    |       |     |
| <b>1a</b> | Distinguish between following giving suitable examples<br>i. Periodic and Aperiodic signal<br>ii. Causal and Non-causal system                                                                                                                                 | 5     | 1   |
| <b>1b</b> | If $x(t) = e^{\frac{j5\pi}{8}t}$ is sampled with sampling interval $T_s = \frac{2}{3}$ . Is the resulting signal periodic? If so, what is its time period?                                                                                                     | 5     | 1   |
| <b>1c</b> | $x(t) = \begin{cases} 1, & 0 < t < T \\ 0, & \text{otherwise} \end{cases}$ $h(t) = \begin{cases} t, & 0 < t < 2T \\ 0, & \text{otherwise} \end{cases}$ Plot the signals $x(t)$ and $h(t)$ . Also find the signal $y(t) = x(t) * h(t)$ using signal operations. | 5     | 1   |
| <b>Q2</b> | <b>Attempt any 2 parts of the questions below</b>                                                                                                                                                                                                              |       |     |
| <b>2a</b> | Find the Laplace transform and ROC of the following signal<br>$x_1(t) = \begin{cases} 2; & 0 < t < 5 \\ 0; & 5 < t < 10 \\ e^{4t}; & 10 < t \end{cases}$                                                                                                       | 5     | 2   |
| <b>2b</b> | Using Laplace transform calculate $y(t)$ for the given equation<br>$y'' + 3y' + 2y = e^t; \quad y(0) = y'(0) = 0$                                                                                                                                              | 5     | 2   |
| <b>2c</b> | Define transfer function.<br>Also, Using Laplace theorem calculate the transfer function and the impulse response for the system which has<br>Output: $y(t) = 10e^{-t}\cos 4t U(t)$<br>Input: $x(t) = e^{-t} U(t)$                                             | 5     | 2   |
| <b>Q3</b> | <b>Attempt any 2 parts of the questions below</b>                                                                                                                                                                                                              |       |     |
| <b>3a</b> | Develop the series and parallel equivalent circuit of an inductor in s-domain.                                                                                                                                                                                 | 5     | 3   |
| <b>3b</b> | In the following circuit, the capacitor C is initially charged to voltage $V_0$ . Determine the nature of current through the resistance R when the switch is                                                                                                  | 5     | 3   |

| Q. No. | Question                                                                                                                                                                                                                                                                                                                                                                                          | Marks | COs |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|
|        | <p>closed at <math>t=0</math>.</p>                                                                                                                                                                                                                                                                               |       |     |
| 3c     | <p>In the circuit shown below, switch K is changed from position <math>a</math> to <math>b</math> at <math>t=0</math>, steady state condition having been reached before switching. Find the values of <math>i</math>; <math>\frac{di}{dt}</math> and <math>\frac{d^2i}{dt^2}</math> at <math>t = 0+</math></p>  | 5     | 3   |
| Q4     | Attempt any 2 parts of the questions below                                                                                                                                                                                                                                                                                                                                                        |       |     |
| 4a     | <p>Find the z-transform of the following:<br/> <math>x[n] = 7\left(\frac{1}{3}\right)^n u[n] - 6\left(\frac{1}{2}\right)^n u[n]</math> and also determine the ROC</p>                                                                                                                                                                                                                             | 5     | 4   |
| 4b     | <p>Determine the response of the following system using z-transform:<br/> <math>x(n+2) - 3x(n+1) + 2x(n) = \delta(n)</math>. Assume that all the initial conditions are zero.</p>                                                                                                                                                                                                                 | 5     | 4   |
| 4c     | <p>The difference equation of a causal discrete time LTI system is expressed as follows: <math>y(n) + 4y(n-1) = x(n)</math>.<br/> Determine the transfer function of the system. Also, determine the response of the system when subjected to an input of <math>u(n)</math></p>                                                                                                                   | 5     | 4   |
| Q5     | Attempt any 2 parts of the questions below                                                                                                                                                                                                                                                                                                                                                        |       |     |
| 5a     | <p>Develop the relationship between trigonometric Fourier coefficients and exponential Fourier coefficients.</p>                                                                                                                                                                                                                                                                                  | 5     | 5   |
| 5b     | <p>Determine the Fourier series representation for the following signal:</p>                                                                                                                                                                                                                                   | 5     | 5   |
| 5c     | <p><math>f(t) = t(\pi - t)</math>, <math>t \in [0, \pi]</math>. Find the trigonometric Fourier series expansion of <math>f(t)</math></p>                                                                                                                                                                                                                                                          | 5     | 5   |